In [56]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

In [57]:
```
df = pd.read_csv('SuperStoreOrders - SuperStoreOrders.csv')
```

In [58]:
```
df.head()
```

Out[58]:

| | order_id | order_date | ship_date | ship_mode | customer_name | segment | state | country | market | region | ... | category | sub_cat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | AG-2011-2040 | 01/01/2011 | 06/01/2011 | Standard Class | Toby Braunhardt | Consumer | Constantine | Algeria | Africa | Africa | ... | Office Supplies | St |
| **1** | IN-2011-47883 | 01/01/2011 | 08/01/2011 | Standard Class | Joseph Holt | Consumer | New South Wales | NaN | APAC | Oceania | ... | NaN | Su |
| **2** | HU-2011-1220 | 01/01/2011 | 05/01/2011 | Second Class | Annie Thurman | Consumer | Budapest | Hungary | EMEA | EMEA | ... | Office Supplies | St |
| **3** | IT-2011-3647632 | 01/01/2011 | 05/01/2011 | Second Class | Eugene Moren | Home Office | Stockholm | Sweden | NaN | North | ... | Office Supplies | |
| **4** | IN-2011-47883 | 01/01/2011 | 08/01/2011 | Standard Class | Joseph Holt | Consumer | New South Wales | Australia | APAC | Oceania | ... | Furniture | Furni |

5 rows × 21 columns

In [59]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        51290 non-null  object
 1   order_date      51290 non-null  object
 2   ship_date       51290 non-null  object
 3   ship_mode       51290 non-null  object
 4   customer_name   51290 non-null  object
 5   segment         51290 non-null  object
 6   state           51290 non-null  object
 7   country         51289 non-null  object
 8   market          51289 non-null  object
 9   region          51288 non-null  object
 10  product_id      51290 non-null  object
 11  category        51289 non-null  object
 12  sub_category    51290 non-null  object
 13  product_name    51290 non-null  object
 14  sales           51285 non-null  object
 15  quantity        51290 non-null  int64
 16  discount        51290 non-null  float64
 17  profit          51289 non-null  float64
 18  shipping_cost   51289 non-null  float64
 19  order_priority  51289 non-null  object
 20  year            51290 non-null  int64
dtypes: float64(3), int64(2), object(16)
memory usage: 8.2+ MB
```

In [60]:
```
df.describe().T
```

Out[60]:

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **quantity** | 51290.0 | 3.476545 | 2.278766 | 1.000 | 2.00 | 3.00 | 5.00 | 14.000 |
| **discount** | 51290.0 | 0.142908 | 0.212280 | 0.000 | 0.00 | 0.00 | 0.20 | 0.850 |
| **profit** | 51289.0 | 28.636635 | 174.421983 | -6599.978 | 0.00 | 9.24 | 36.81 | 8399.976 |
| **shipping_cost** | 51289.0 | 26.376394 | 57.297260 | 0.000 | 2.61 | 7.79 | 24.45 | 933.570 |
| **year** | 51290.0 | 2012.777208 | 1.098931 | 2011.000 | 2012.00 | 2013.00 | 2014.00 | 2014.000 |

In [61]:
```
df.dtypes
```

Out[61]:
```
order_id           object
order_date         object
ship_date          object
ship_mode          object
customer_name      object
segment            object
state              object
country            object
market             object
region             object
product_id         object
category           object
sub_category       object
product_name       object
sales              object
quantity            int64
discount          float64
profit            float64
shipping_cost     float64
order_priority     object
year                int64
dtype: object
```

In [62]:
```
df.shape
```

Out[62]:
```
(51290, 21)
```

# data cleaning

In [63]:
```
df.isnull().sum()
```

Out[63]:
```
order_id          0
order_date        0
ship_date         0
ship_mode         0
customer_name     0
segment           0
state             0
country           1
market            1
region            2
product_id        0
category          1
sub_category      0
product_name      0
sales             5
quantity          0
discount          0
profit            1
shipping_cost     1
order_priority    1
year              0
dtype: int64
```

In [65]:
```
df.dtypes
```

Out[65]:
```
order_id                  object
order_date        datetime64[ns]
ship_date         datetime64[ns]
ship_mode                 object
customer_name             object
segment                   object
state                     object
country                   object
market                    object
region                    object
product_id                object
category                  object
sub_category              object
product_name              object
sales                     object
quantity                   int64
discount                 float64
profit                   float64
shipping_cost            float64
order_priority            object
year                       int64
dtype: object
```

In [66]:
```
df['order_date'] = pd.to_datetime(df['order_date'], dayfirst=True)
df['ship_date']  = pd.to_datetime(df['ship_date'],  dayfirst=True)
```

In [67]:
```
df['sales'] = df['sales'].astype(str).str.replace(',','').str.strip()
```

In [68]:
```
df['sales'] = pd.to_numeric(df['sales'] , errors='coerce')
```

In [69]:
```
df['sales'].isna().sum()
```

Out[69]:
```
np.int64(5)
```

In [70]:
```
df.isna().sum()
```

Out[70]:
```
order_id          0
order_date        0
ship_date         0
ship_mode         0
customer_name     0
segment           0
state             0
country           1
market            1
region            2
product_id        0
category          1
sub_category      0
product_name      0
sales             5
quantity          0
discount          0
profit            1
shipping_cost     1
order_priority    1
year              0
dtype: int64
```

In [71]:
```
df['country'].mode()[0]
```

Out[71]:
```
'United States'
```

In [78]:
```
df['country'] = df['country'].fillna(df['country'].mode()[0] )
df['market']  = df['market'].fillna(df['market'].mode()[0] )
df['region']  = df['region'].fillna(df['region'].mode()[0] )
df['category']= df['category'].fillna(df['category'].mode()[0])
df['order_priority'] = df['order_priority'].fillna(df['order_priority'].mode()[0])
```

In [76]:
```
df['sales'] = df.groupby('product_name')['sales'].transform(lambda x:x.fillna(x.mean()))
df['profit'] = df.groupby('product_name')['profit'].transform(lambda x:x.fillna(x.mean()))
df['shipping_cost'] = df.groupby('product_name')['shipping_cost'].transform(lambda x:x.fillna(x.mean()))
```

In [79]:
```
df.isna().sum()
```

Out[79]:
```
order_id          0
order_date        0
ship_date         0
ship_mode         0
customer_name     0
segment           0
state             0
country           0
market            0
region            0
product_id        0
category          0
sub_category      0
product_name      0
sales             0
quantity          0
discount          0
profit            0
shipping_cost     0
order_priority    0
year              0
dtype: int64
```

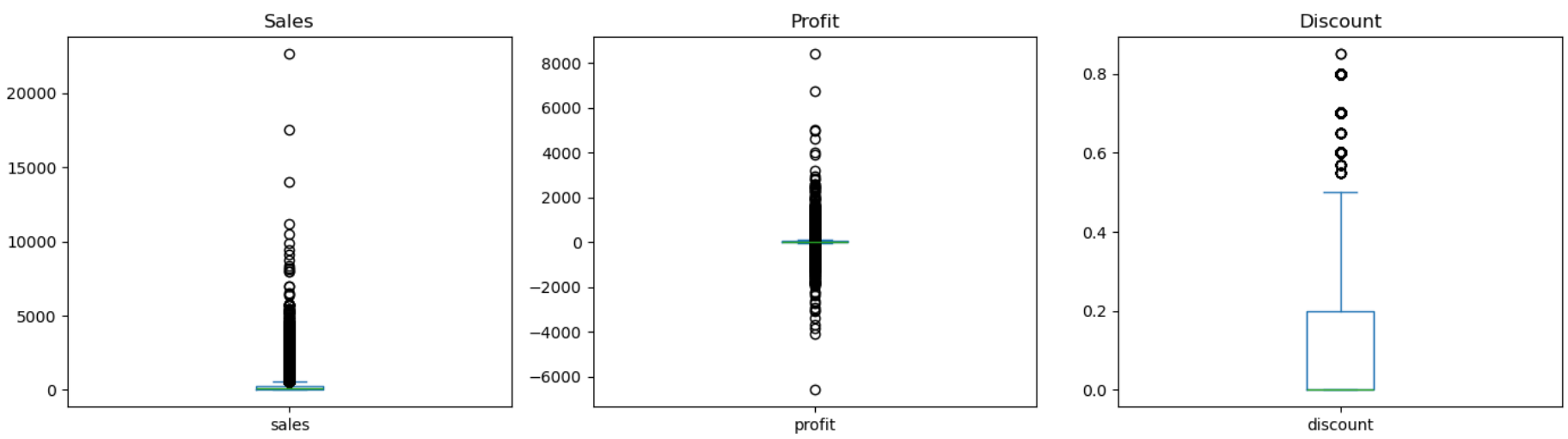
In [80]:
```
df.dtypes
```

Out[80]:
```
order_id                  object
order_date        datetime64[ns]
ship_date         datetime64[ns]
ship_mode                 object
customer_name             object
segment                   object
state                     object
country                   object
market                    object
region                    object
product_id                object
category                  object
sub_category              object
product_name              object
sales                    float64
quantity                   int64
discount                 float64
profit                   float64
shipping_cost            float64
order_priority            object
year                       int64
dtype: object
```

In [81]:
```
df.duplicated().sum()
```

Out[81]:
```
np.int64(0)
```

In [83]:
```
fig, axes = plt.subplots(1, 3, figsize=(14, 4))

df['sales'].plot(kind='box', ax=axes[0], title='Sales')
df['profit'].plot(kind='box', ax=axes[1], title='Profit')
df['discount'].plot(kind='box', ax=axes[2], title='Discount')

plt.tight_layout()
plt.show()
```




In [82]:
```
df[['sales', 'profit', 'discount', 'quantity', 'shipping_cost']].describe()
```

Out[82]:

| | sales | profit | discount | quantity | shipping_cost |
|---|---|---|---|---|---|
| **count** | 51290.000000 | 51290.000000 | 51290.000000 | 51290.000000 | 51290.000000 |
| **mean** | 246.472835 | 28.637459 | 0.142908 | 3.476545 | 26.376092 |
| **std** | 487.499722 | 174.420382 | 0.212280 | 2.278766 | 57.296742 |
| **min** | 0.000000 | -6599.978000 | 0.000000 | 1.000000 | 0.000000 |
| **25%** | 31.000000 | 0.000000 | 0.000000 | 2.000000 | 2.610000 |
| **50%** | 85.000000 | 9.240000 | 0.000000 | 3.000000 | 7.790000 |
| **75%** | 251.000000 | 36.810000 | 0.200000 | 5.000000 | 24.450000 |
| **max** | 22638.000000 | 8399.976000 | 0.850000 | 14.000000 | 933.570000 |

In [85]:
```
df[df['profit'] < 0]
```

Out[85]:

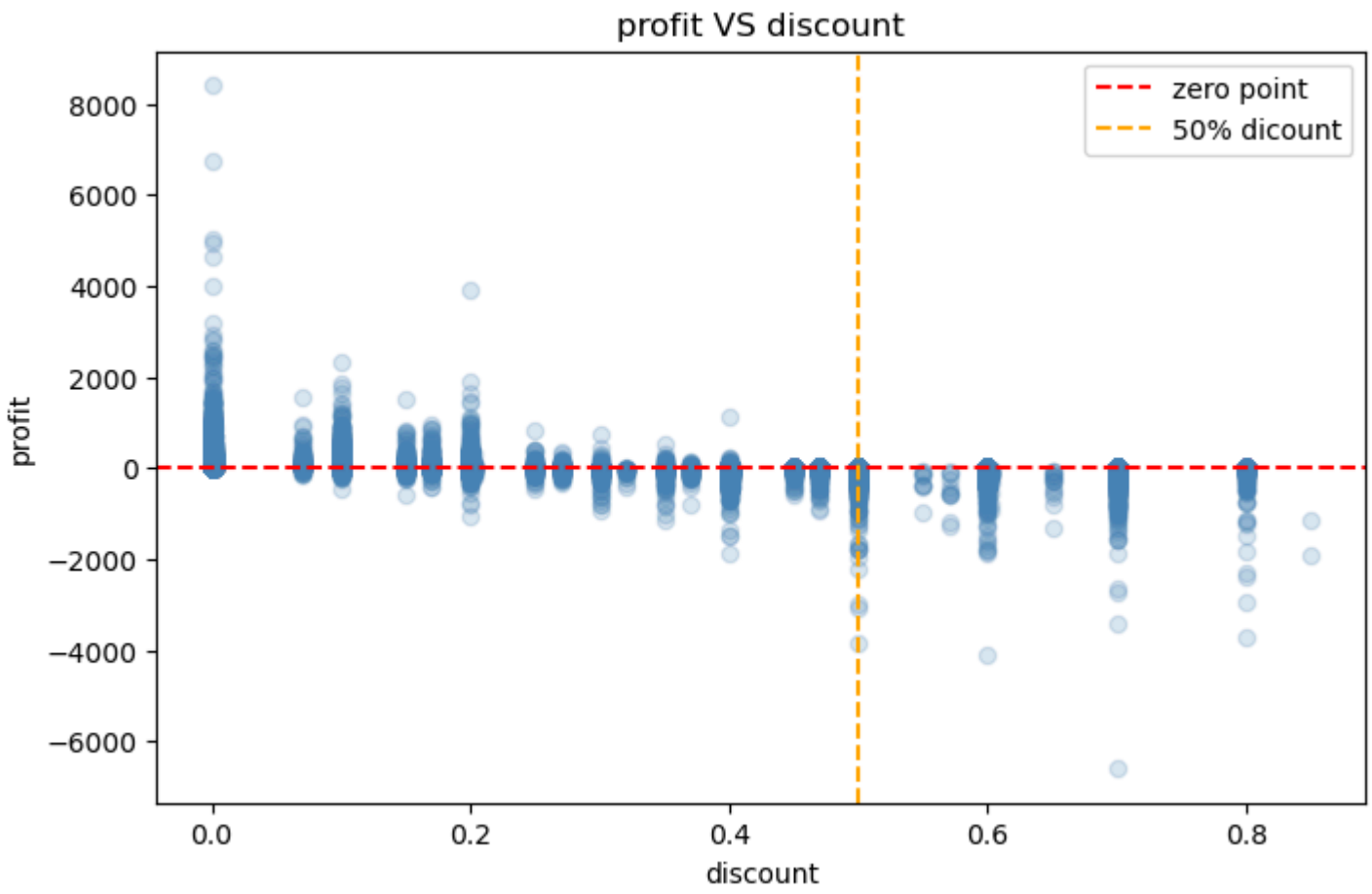
| | order_id | order_date | ship_date | ship_mode | customer_name | segment | state | country | market | region | ... | category |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **3** | IT-2011-3647632 | 2011-01-01 | 2011-01-05 | Second Class | Eugene Moren | Home Office | Stockholm | Sweden | APAC | North | ... | Office Supplies |
| **8** | ID-2011-80230 | 2011-01-03 | 2011-01-09 | Standard Class | Ken Lonsdale | Consumer | Auckland | New Zealand | APAC | Central | ... | Technology |
| **10** | IN-2011-65159 | 2011-01-03 | 2011-01-07 | Second Class | Larry Blacks | Consumer | National Capital | Philippines | APAC | Southeast Asia | ... | Office Supplies |
| **11** | IN-2011-65159 | 2011-01-03 | 2011-01-07 | Second Class | Larry Blacks | Consumer | National Capital | Philippines | APAC | Southeast Asia | ... | Furniture |
| **14** | ID-2011-80230 | 2011-01-03 | 2011-01-09 | Standard Class | Ken Lonsdale | Consumer | Auckland | New Zealand | APAC | Oceania | ... | Furniture |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **51267** | IT-2014-3715679 | 2014-12-31 | 2015-01-05 | Second Class | Jim Kriz | Home Office | North Brabant | Netherlands | EU | Central | ... | Office Supplies |
| **51272** | IT-2014-3715679 | 2014-12-31 | 2015-01-05 | Second Class | Jim Kriz | Home Office | North Brabant | Netherlands | EU | Central | ... | Office Supplies |
| **51274** | TU-2014-5170 | 2014-12-31 | 2015-01-04 | Second Class | Valerie Dominguez | Consumer | Konya | Turkey | EMEA | EMEA | ... | Technology |
| **51275** | ID-2014-54442 | 2014-12-31 | 2015-01-04 | Standard Class | Muhammed Yedwab | Corporate | National Capital | Philippines | APAC | Southeast Asia | ... | Technology |
| **51289** | CA-2014-156720 | 2014-12-31 | 2015-01-04 | Standard Class | Jill Matthias | Consumer | Colorado | United States | US | West | ... | Office Supplies |

12543 rows × 21 columns

In [88]:
```
df['discount'].corr(df['profit']) # العلاقه بين المكسب والخصم عكسيه كل ما الخصم بيزيد المكسب بيزيد ودا بيوضح ليه في مكسب بالسالب
```

Out[88]:
```
np.float64(-0.3163652018750581)
```

In [91]:
```
import matplotlib.pyplot as plt
plt.figure(figsize=(8, 5))
plt.scatter(df['discount'], df['profit'], alpha=0.2, color='steelblue')
plt.axhline(0, color='red', linestyle='--', label='zero point')
plt.axvline(0.5, color='orange', linestyle='--', label='50% dicount')
plt.xlabel('discount')
plt.ylabel('profit')
plt.title('profit VS discount')
plt.legend()
plt.show()
```




# EDA

## Profit & Loss

### What is the total profit by category?

In [96]:
```
df.groupby('category').agg(
    total_gsins = ('profit' , lambda x:x[x>0].sum()),
    total_losses = ('profit' , lambda x:x[x<0].sum()),
    total_Profit = ('profit' , 'sum')
).round(2)
```

Out[96]:

| category | total_gsins | total_losses | total_Profit |
|---|---|---|---|
| **Furniture** | 656442.78 | -369880.07 | 286562.71 |
| **Office Supplies** | 782455.75 | -263981.92 | 518473.83 |
| **Technology** | 950274.13 | -286495.40 | 663778.73 |

### Which sub-category has the highest profit?

In [97]:
```
df.groupby('sub_category').agg(
    total_gsins = ('profit' , lambda x:x[x>0].sum()),
    total_losses = ('profit' , lambda x:x[x<0].sum()),
    total_Profit = ('profit' , 'sum')
).round(2).head(5)
```

Out[97]:

| sub_category | total_gsins | total_losses | total_Profit |
|---|---|---|---|
| **Accessories** | 169483.80 | -39857.50 | 129626.31 |
| **Appliances** | 205672.28 | -63991.69 | 141680.59 |
| **Art** | 79614.54 | -21660.63 | 57953.91 |
| **Binders** | 125333.91 | -52884.06 | 72449.85 |
| **Bookcases** | 263151.17 | -101446.30 | 161704.88 |

### Which country has the highest profit?

In [118…
```
df.groupby('country').agg(
    total_gsins = ('profit' , lambda x:x[x>0].sum()),
    total_losses = ('profit' , lambda x:x[x<0].sum()),
    total_Profit = ('profit' , 'sum')
).round(2).head(10)
```

Out[118…

| country | total_gsins | total_losses | total_Profit |
|---|---|---|---|
| **Afghanistan** | 5460.30 | 0.00 | 5460.30 |
| **Albania** | 709.32 | 0.00 | 709.32 |
| **Algeria** | 9106.50 | 0.00 | 9106.50 |
| **Angola** | 6494.97 | 0.00 | 6494.97 |
| **Argentina** | 781.85 | -19475.65 | -18693.80 |
| **Armenia** | 69.09 | 0.00 | 69.09 |
| **Australia** | 146067.91 | -40618.99 | 105448.93 |
| **Austria** | 24341.70 | 0.00 | 24341.70 |
| **Azerbaijan** | 1831.05 | 0.00 | 1831.05 |
| **Bahrain** | 12.84 | 0.00 | 12.84 |

### What percentage of orders are unprofitable?

In [103…

```
(df['profit']<0).mean().round(2)*100
```

Out[103…

```
np.float64(24.0)
```

### What is the overall profit margin?

In [105…

```
((df['profit'].sum() ) / (df['sales'].sum()) *100).round(2)
```

Out[105…

```
np.float64(11.62)
```

## Discount Impact

### At what discount rate do we start losing money?

In [110…

```
df.groupby(pd.cut(df['discount'], bins=[0, 0.1, 0.2, 0.3, 0.5, 0.8, 1.0]) ,observed= True)['profit'].mean().round(2)
```

Out[110…

```
discount
(0.0, 0.1]      72.62
(0.1, 0.2]      27.61
(0.2, 0.3]     -21.88
(0.3, 0.5]     -61.55
(0.5, 0.8]     -98.20
(0.8, 1.0]   -1534.33
Name: profit, dtype: float64
```

### How strong is the correlation between discount and profit?

In [112…

```
df['discount'].corr(df['profit']).round(2)
```

Out[112…

```
np.float64(-0.32)
```

### Which category is most affected by discounts?

In [116…

```
df.groupby('category')[['discount', 'profit']].corr().unstack()['discount']['profit'].round(2)
```

Out[116…

```
category
Furniture         -0.49
Office Supplies   -0.25
Technology        -0.38
Name: profit, dtype: float64
```

## Sales Performance

### sales trend over year

In [121…

```
df.groupby('year')['sales'].sum()
```

Out[121…

```
year
2011   2,258,198
2012   2,677,493
2013   3,405,860
2014   4,300,041
Name: sales, dtype: float64
```

### Which market has the highest sales and profit?

In [122…

```
df.groupby('market').agg(
    sales = ('sales' , 'sum'),
    profit = ('profit' , 'sum')
).sort_values('profit' , ascending= False)
```

Out[122…

| market | sales | profit |
|---|---|---|
| **APAC** | 3,584,527 | 437,552 |
| **EU** | 2,938,094 | 372,636 |
| **US** | 2,297,354 | 286,397 |
| **LATAM** | 2,164,673 | 221,643 |
| **Africa** | 783,802 | 88,872 |
| **EMEA** | 806,209 | 43,898 |
| **Canada** | 66,932 | 17,817 |

### Which country generates the most sales?

In [127…

```
df.groupby('country')['sales'].sum().nlargest(10)
```

Out[127…

```
country
United States    2,297,474
Australia          925,137
France             858,930
China              699,240
Germany            628,857
Mexico             622,620
India              589,664
United Kingdom     528,570
Indonesia          404,887
Brazil             361,098
Name: sales, dtype: float64
```

### Which region performs best?

In [129…

```
df.groupby('region').agg(
    sales = ('sales' , 'sum'),
    profit = ('profit' , 'sum'),
    orders = ('order_id' , 'count')
).sort_values('profit' , ascending= False)
```

Out[129…

| region | sales | profit | orders |
|---|---|---|---|
| **Central** | 2,823,366 | 311,088 | 11119 |
| **North** | 1,248,192 | 194,378 | 4785 |
| **North Asia** | 846,998 | 165,578 | 2338 |
| **South** | 1,600,960 | 140,356 | 6645 |
| **Central Asia** | 752,839 | 132,480 | 2048 |
| **Oceania** | 1,099,226 | 121,983 | 3485 |
| **West** | 725,514 | 108,418 | 3203 |
| **East** | 678,834 | 91,523 | 2848 |
| **Africa** | 783,802 | 88,872 | 4587 |
| **EMEA** | 806,209 | 43,898 | 5029 |
| **Caribbean** | 324,281 | 34,571 | 1690 |
| **Southeast Asia** | 884,438 | 17,852 | 3129 |
| **Canada** | 66,932 | 17,817 | 384 |

## Customer Behavior

### Who are the top 10 customers by sales?

In [131…

```
df.groupby('customer_name')['sales'].sum().nlargest(10)
```

Out[131…
```
customer_name
Tom Ashbrook         40,489
Tamara Chand         37,453
Greg Tran            35,552
Christopher Conant   35,187
Sean Miller          35,170
Bart Watters         32,315
Natalie Fritzler     31,778
Fred Hopkins         30,404
Jane Waco            30,288
Hunter Lopez         30,246
Name: sales, dtype: float64
```

### Who are the top 10 customers by profit?

In [132…
```
df.groupby('customer_name')['profit'].sum().nlargest(10)
```

Out[132…
```
customer_name
Tamara Chand      8,673
Raymond Buch      8,453
Sanjit Chand      8,205
Hunter Lopez      7,817
Bill Eplett       7,410
Harry Marie       6,958
Susan Pistek      6,484
Mike Gockenbach   6,459
Adrian Barton     6,417
Tom Ashbrook      6,312
Name: profit, dtype: float64
```

### How many orders does each segment place?

In [136…
```
df.groupby('segment')['order_id'].count()
```

Out[136…
```
segment
Consumer       26518
Corporate      15429
Home Office     9343
Name: order_id, dtype: int64
```

### Which customer segment is most profitable?

In [137…
```
df.groupby('segment')['profit'].sum().sort_values(ascending= False)
```

Out[137…
```
segment
Consumer      749,020
Corporate     442,786
Home Office   277,009
Name: profit, dtype: float64
```

## Operations & Shipping

### What is the average shipping time per ship mode?

In [138…
```
df['days_to_ship'] = (df['ship_date'] - df['order_date']).dt.days
```

In [141…
```
df.groupby('ship_mode')['days_to_ship'].mean().sort_values(ascending= False)
```

Out[141…
```
ship_mode
Standard Class   5
Second Class     3
First Class      2
Same Day         0
Name: days_to_ship, dtype: float64
```

### Does order priority actually affect shipping speed?

In [142…
```
df.groupby('order_priority')['days_to_ship'].mean()
```

Out[142…
```
order_priority
Critical   2
High       3
Low        6
Medium     5
Name: days_to_ship, dtype: float64
```

### What is the average shipping cost per ship mode?

In [143…
```
df.groupby('ship_mode')['shipping_cost'].mean()
```

Out[143…

```
ship_mode
First Class      41
Same Day         43
Second Class     30
Standard Class   20
Name: shipping_cost, dtype: float64
```

### Does shipping cost negatively impact profit?

In [147…

```
df['shipping_cost'].corr(df['profit']).round(2) # So shipping cost is not causing the profit to increase
# large orders naturally come with both higher profit and higher shipping cost together.
```

Out[147…

```
np.float64(0.36)
```

## Products

### What are the top 10 products by quantity sold?

In [148…

```
df.groupby('product_name')['quantity'].sum().nlargest(10)
```

Out[148…

```
product_name
Staples                                   876
Cardinal Index Tab, Clear                 337
Eldon File Cart, Single Width             321
Rogers File Cart, Single Width            262
Sanford Pencil Sharpener, Water Color     259
Stockwell Paper Clips, Assorted Sizes     253
Avery Index Tab, Clear                    252
Ibico Index Tab, Clear                    251
Smead File Cart, Single Width             250
Stanley Pencil Sharpener, Water Color     242
Name: quantity, dtype: int64
```

### What are the top 10 products by sales value?

In [149…

```
df.groupby('product_name')['sales'].sum().nlargest(10)
```

Out[149…

```
product_name
Apple Smart Phone, Full Size                              86,936
Cisco Smart Phone, Full Size                              76,441
Motorola Smart Phone, Full Size                           73,159
Nokia Smart Phone, Full Size                              71,904
Canon imageCLASS 2200 Advanced Copier                     61,600
Hon Executive Leather Armchair, Adjustable                58,200
Office Star Executive Leather Armchair, Adjustable        50,667
Harbour Creations Executive Leather Armchair, Adjustable  50,120
Samsung Smart Phone, Cordless                             48,654
Nokia Smart Phone, with Caller ID                         47,880
Name: sales, dtype: float64
```

## Time Analysis

### What are the monthly sales trends?

In [150…

```
df['month'] = df['order_date'].dt.month
df.groupby('month')['sales'].sum()
```

Out[150…

```
month
1       673,828
2       543,768
3       770,519
4       698,603
5       904,061
6     1,269,751
7       749,423
8     1,293,852
9     1,437,432
10    1,168,220
11    1,551,319
12    1,580,816
Name: sales, dtype: float64
```

### Which month has the highest sales?

In [151…

```
df.groupby('month')['sales'].sum().idxmax()
```

Out[151…

```
np.int32(12)
```

### Which month has the highest profit?

In [152…

```
df.groupby('month')['profit'].sum().idxmax()
```

Out[152… np.int32(11)

### What is the year-over-year sales growth?

In [153…
```
df.groupby('year')['sales'].sum().pct_change() * 100
```

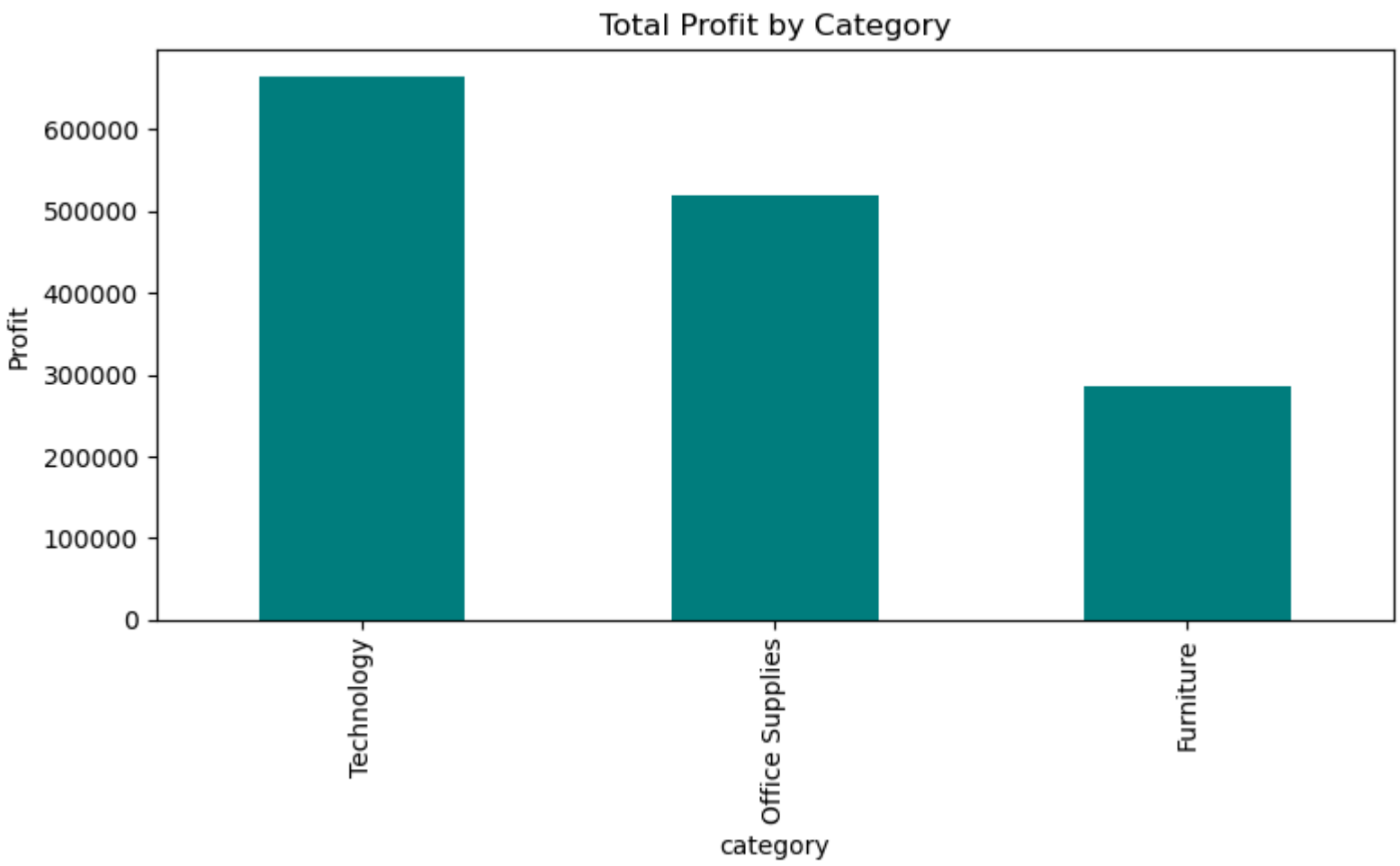
Out[153…
```
year
2011   NaN
2012    19
2013    27
2014    26
Name: sales, dtype: float64
```

In [ ]:

## Data Visu

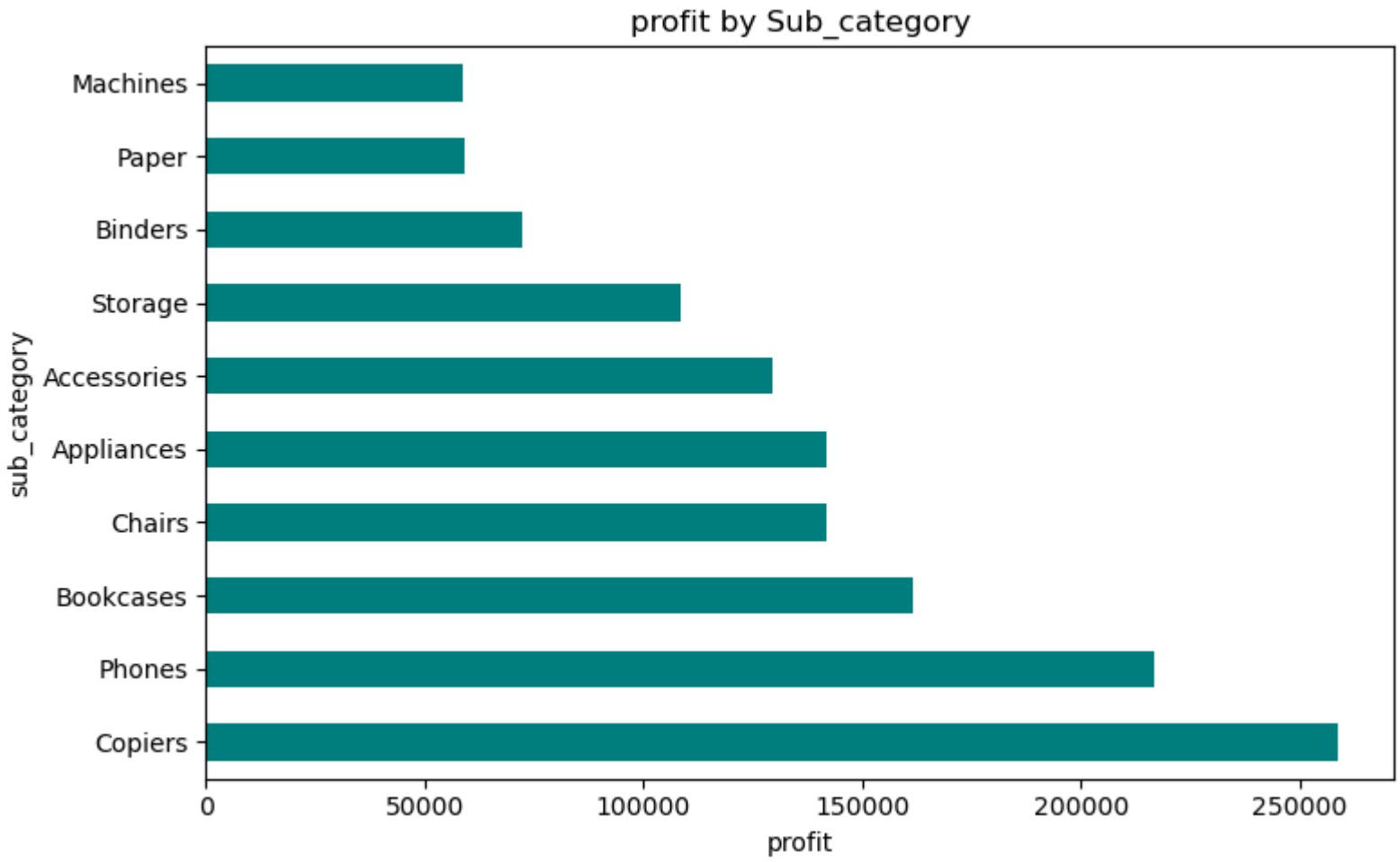
### Total Profit by Category

In [164…
```
profit_cat = df.groupby('category')['profit'].sum().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
profit_cat.plot(kind='bar', color='teal')
plt.title('Total Profit by Category')
plt.ylabel('Profit')
plt.tight_layout()
plt.show()
```




### Total profit by sub_category

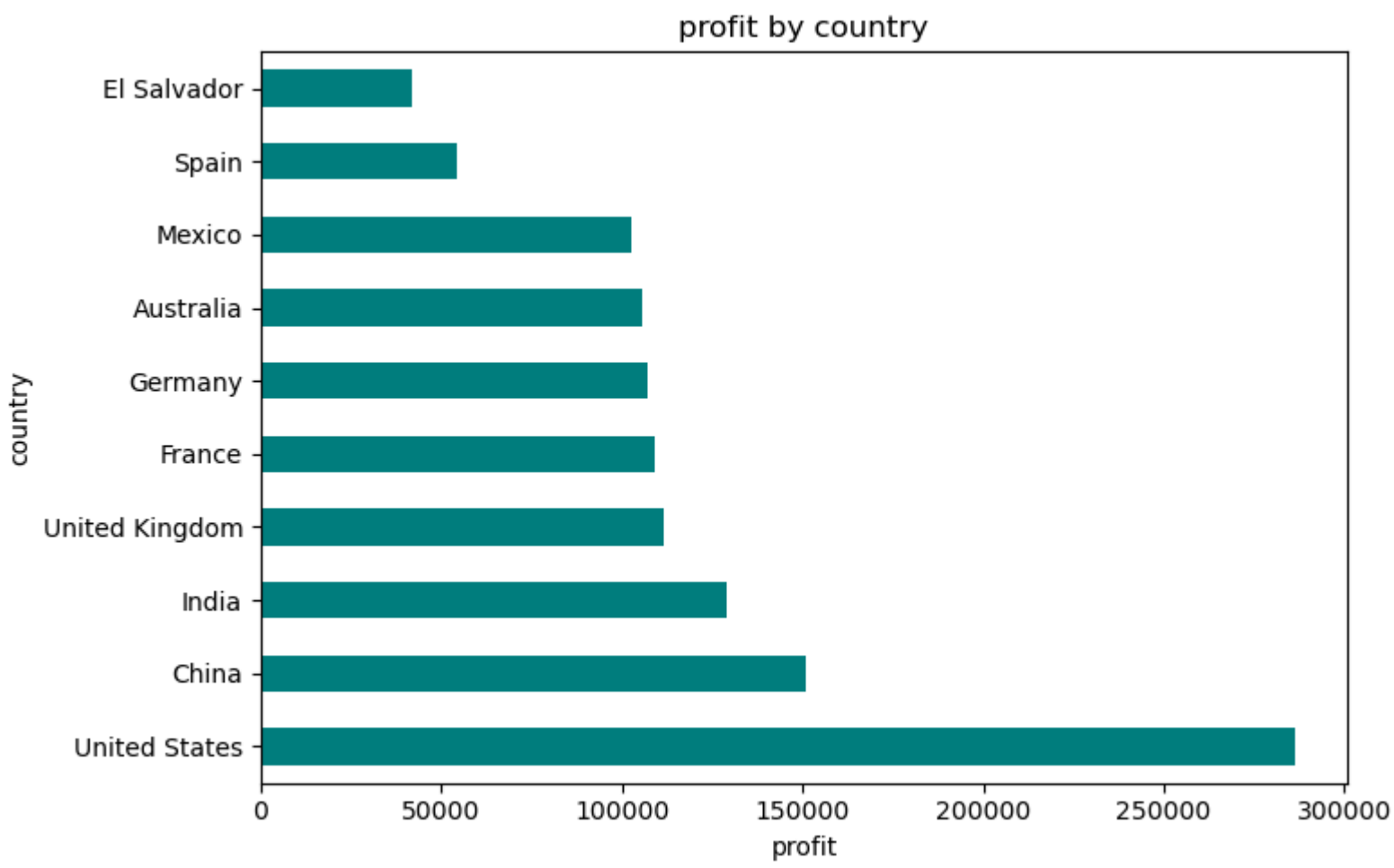
In [172…
```
sub_cat_profit = df.groupby('sub_category')['profit'].sum().sort_values(ascending= False).nlargest(10)
plt.figure(figsize= (8,5))
sub_cat_profit.plot(kind='barh' , color = 'teal')
plt.title('profit by Sub_category')
plt.ylabel('sub_category')
plt.xlabel('profit')
plt.tight_layout()
plt.show()
```




### Which country has the highest profit?

In [174…

```
sub_cat_profit = df.groupby('country')['profit'].sum().sort_values(ascending= False).nlargest(10)
plt.figure(figsize= (8,5))
sub_cat_profit.plot(kind='barh' , color = 'teal')
plt.title('profit by country')
plt.ylabel('country')
plt.xlabel('profit')
plt.tight_layout()
plt.show()
```




In [176…

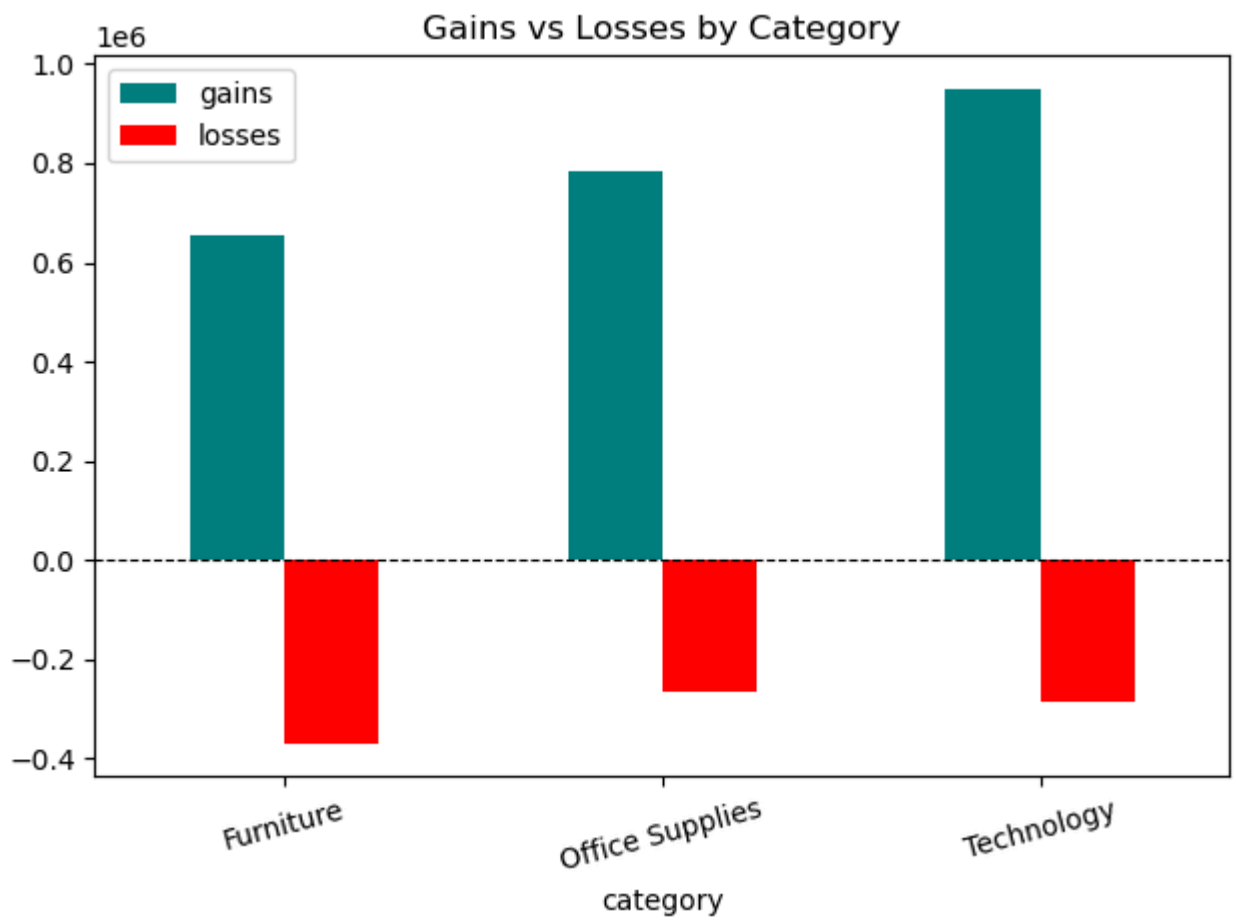
```
import plotly.express as p
top10_countries = df.groupby('country')['profit'].sum().nlargest(10).reset_index()
top10_countries.columns = ['country', 'profit']

fig = px.choropleth(
    top10_countries,
    locations='country',
    locationmode='country names',
    color='profit',
    title='Top 10 Countries by Profit',
    color_continuous_scale='teal'
)

fig.show()
```

### Gains vs Losses by Category

In [182…

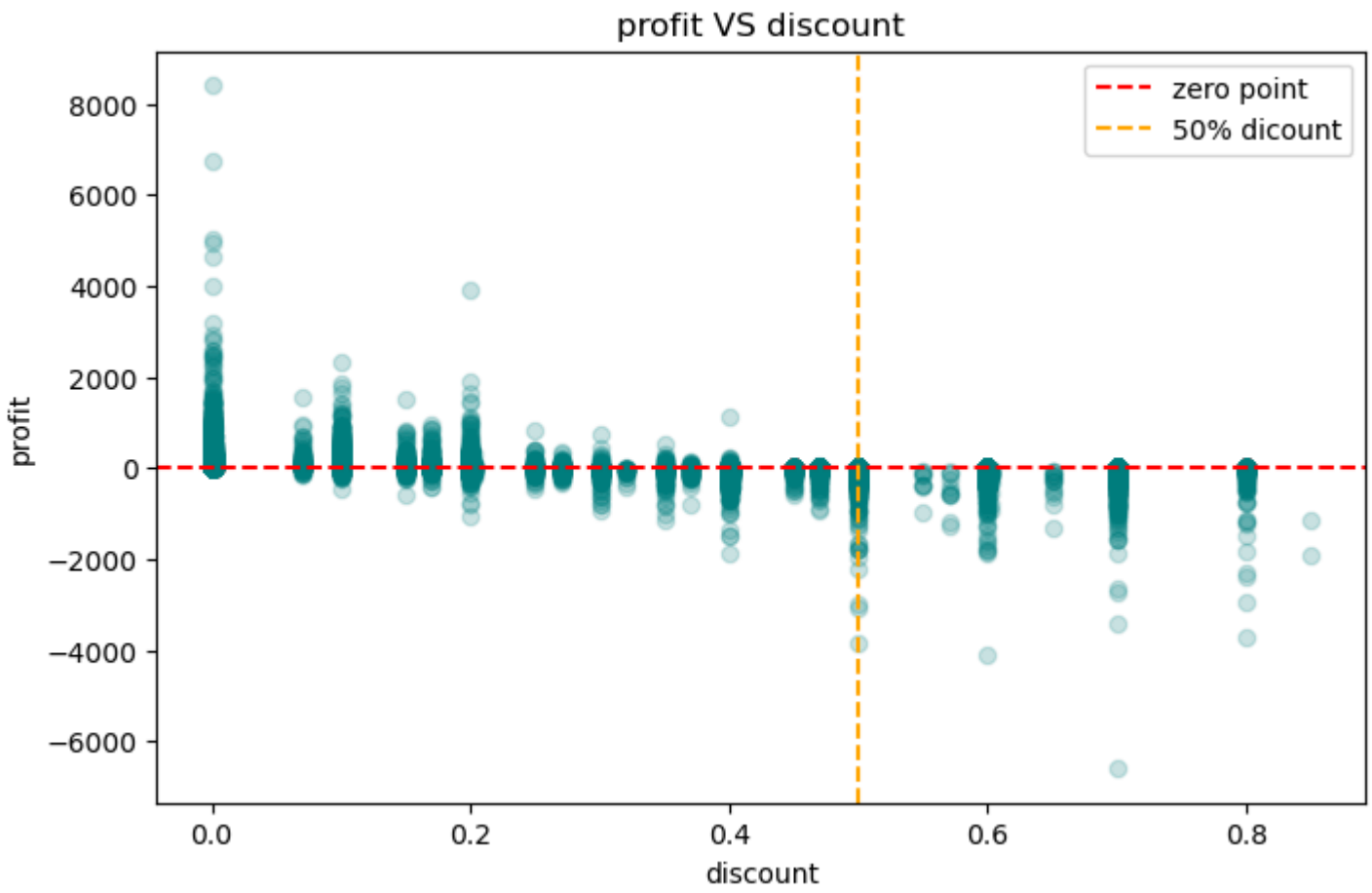
```
profit_detail = df.groupby('category').agg(
    gains  = ('profit', lambda x: x[x > 0].sum()),
    losses = ('profit', lambda x: x[x < 0].sum())
)
plt.figure(figsize=(8, 5))
profit_detail.plot(kind='bar', color=['teal', 'red'])
plt.axhline(0, color='black', linestyle='--', linewidth=0.8)
plt.title('Gains vs Losses by Category')
plt.tick_params(axis='x', rotation=15)
plt.tight_layout()
plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```




In [183…

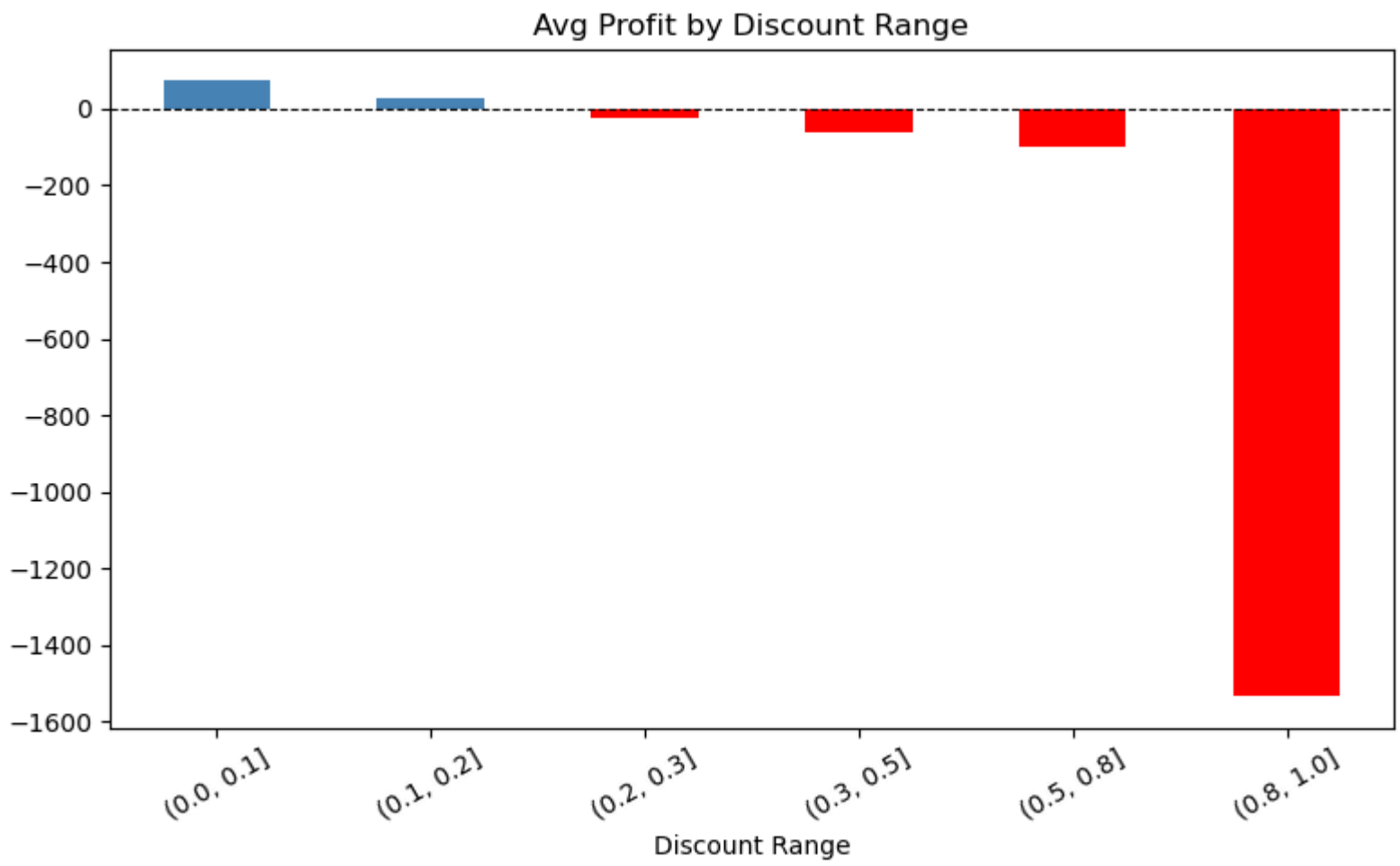
```
plt.figure(figsize=(8, 5))
plt.scatter(df['discount'], df['profit'], alpha=0.2, color='teal')
plt.axhline(0, color='red', linestyle='--', label='zero point')
plt.axvline(0.5, color='orange', linestyle='--', label='50% dicount')
plt.xlabel('discount')
plt.ylabel('profit')
plt.title('profit VS discount')
plt.legend()
plt.show()
```




### Avg Profit by Discount Range

In [185…

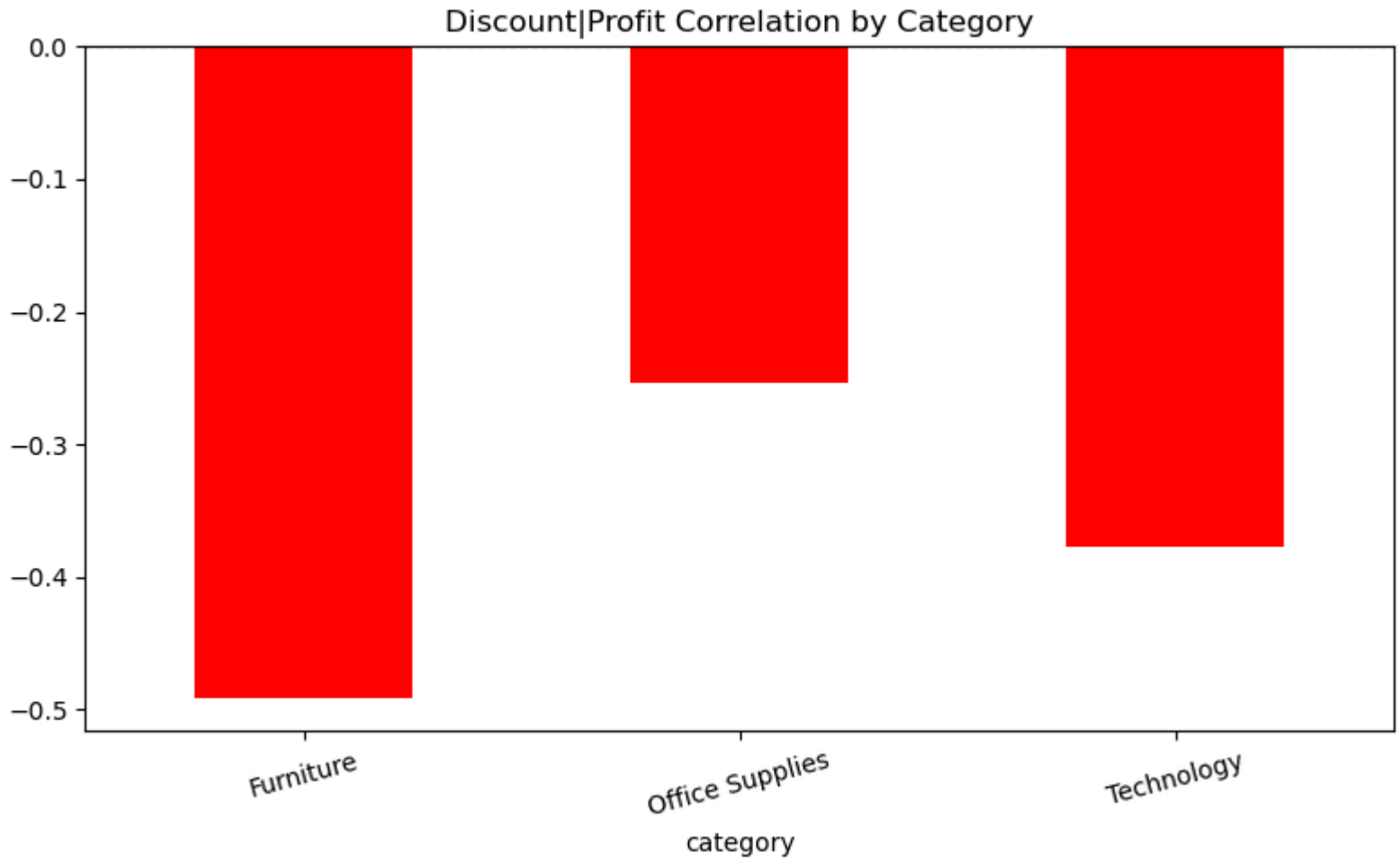
```
disc_bins = df.groupby(
    pd.cut(df['discount'], bins=[0, 0.1, 0.2, 0.3, 0.5, 0.8, 1.0]),
    observed=True
)['profit'].mean()
colors3 = ['red' if x < 0 else 'steelblue' for x in disc_bins]
plt.figure(figsize=(8, 5))
disc_bins.plot(kind='bar', color=colors3)
plt.axhline(0, color='black', linestyle='--', linewidth=0.8)
plt.title('Avg Profit by Discount Range')
plt.xlabel('Discount Range')
plt.tick_params(axis='x', rotation=30)
plt.tight_layout()
plt.show()
```




### Discount-Profit Correlation by Category

In [188…

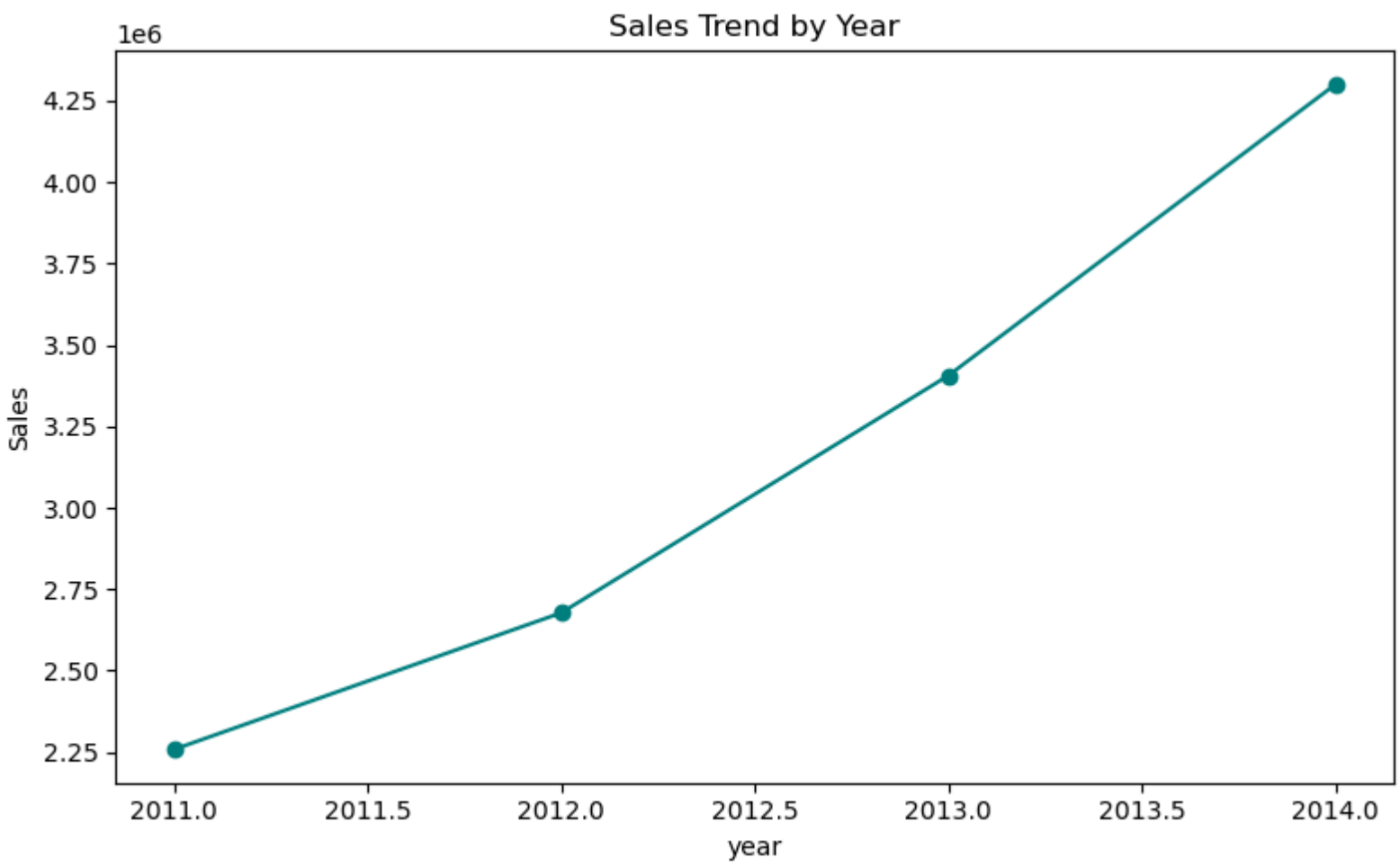
```
corr_cat = df.groupby('category')[['discount', 'profit']].corr().unstack()['discount']['profit']
colors4 = ['red' if x < 0 else 'steelblue' for x in corr_cat]
plt.figure(figsize=(8, 5))
corr_cat.plot(kind='bar', color=colors4)
plt.axhline(0, color='black', linestyle='--', linewidth=0.8)
plt.title('Discount|Profit Correlation by Category')
plt.tick_params(axis='x', rotation=15)
plt.tight_layout()
plt.show()
```




### Sales Trend by Year

In [189…

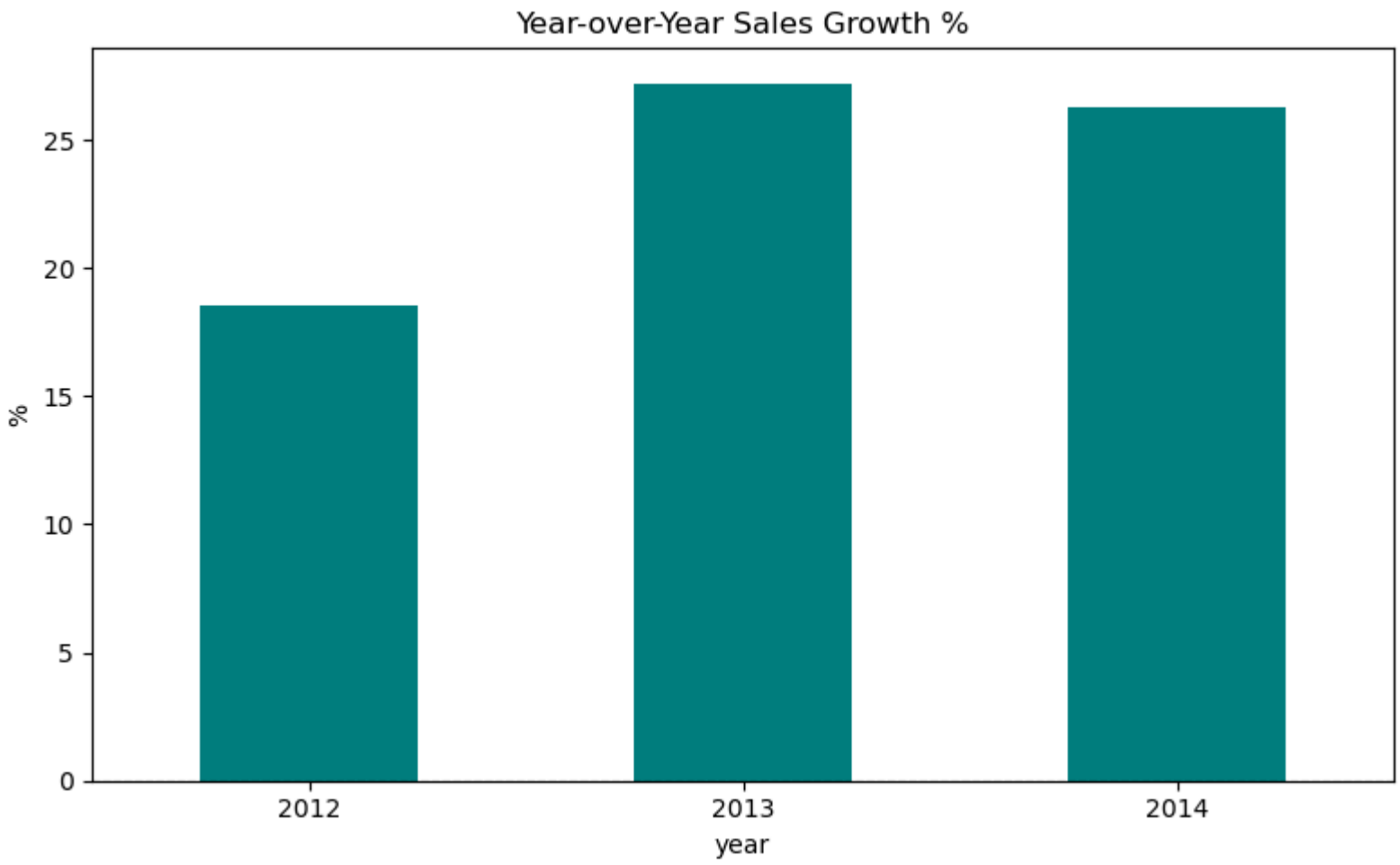
```
sales_year = df.groupby('year')['sales'].sum()
plt.figure(figsize=(8, 5))
sales_year.plot(kind='line', marker='o', color='teal')
plt.title('Sales Trend by Year')
plt.ylabel('Sales')
plt.tight_layout()
plt.show()
```




### Year-over-Year Sales Growth %

In [190…

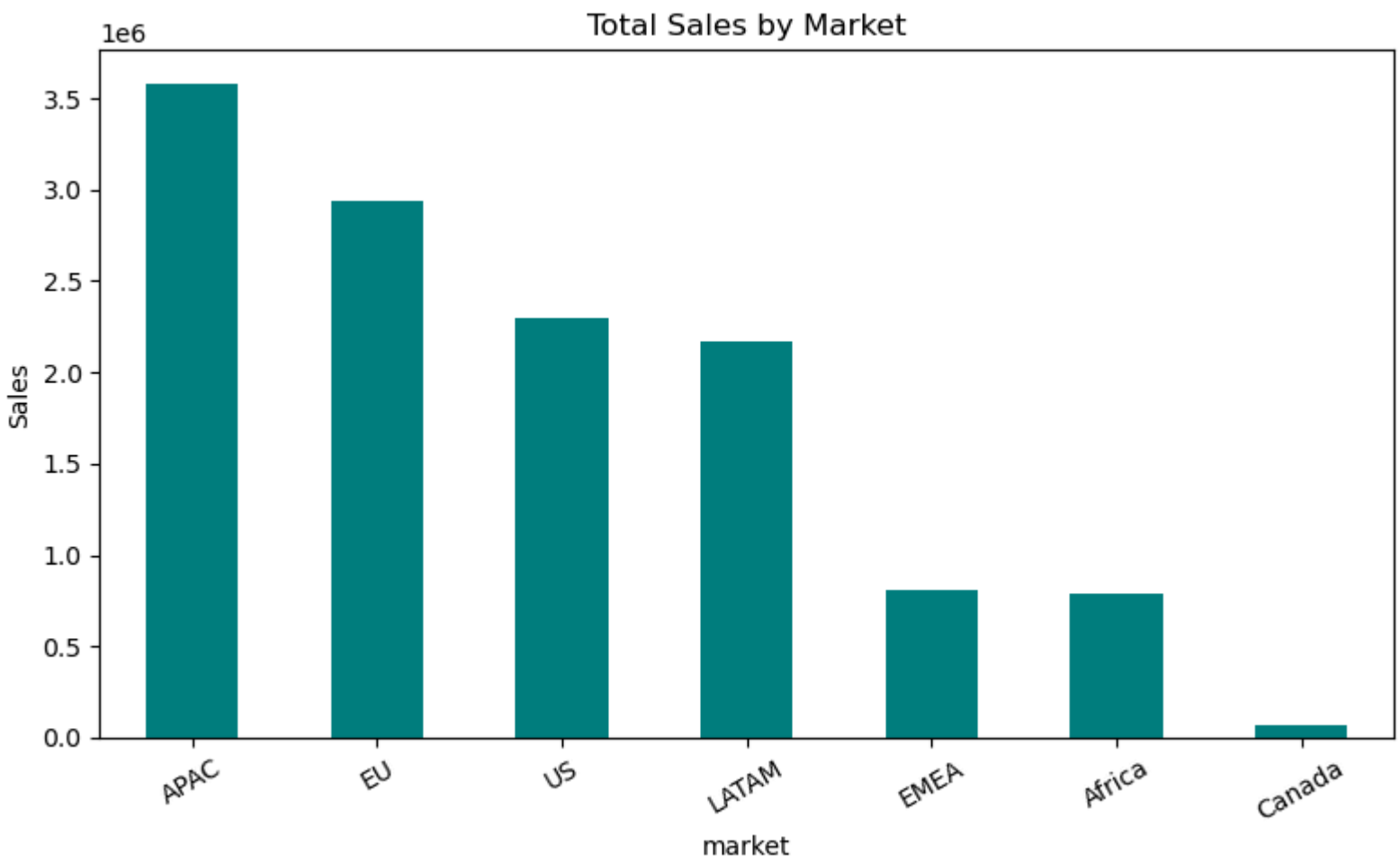
```
growth = df.groupby('year')['sales'].sum().pct_change() * 100
plt.figure(figsize=(8, 5))
growth.dropna().plot(kind='bar', color='teal')
plt.axhline(0, color='black', linestyle='--', linewidth=0.8)
plt.title('Year-over-Year Sales Growth %')
plt.ylabel('%')
plt.tick_params(axis='x', rotation=0)
plt.tight_layout()
plt.show()
```




### Total Sales by Market

In [192…

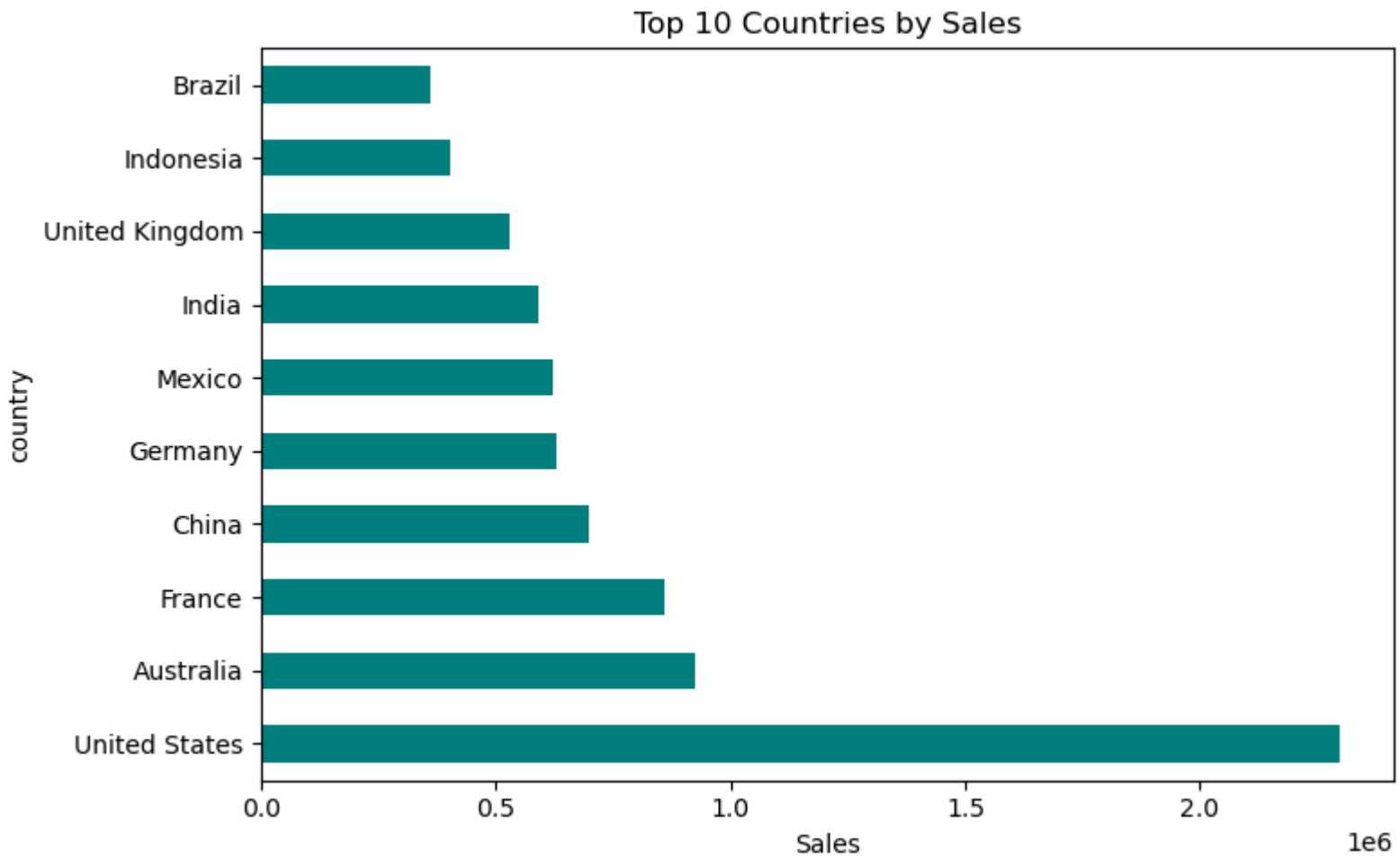
```
market_sales = df.groupby('market')['sales'].sum().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
market_sales.plot(kind='bar', color='teal')
plt.title('Total Sales by Market')
plt.ylabel('Sales')
plt.tick_params(axis='x', rotation=30)
plt.tight_layout()
plt.show()
```




### Top 10 Countries by Sales

In [201…

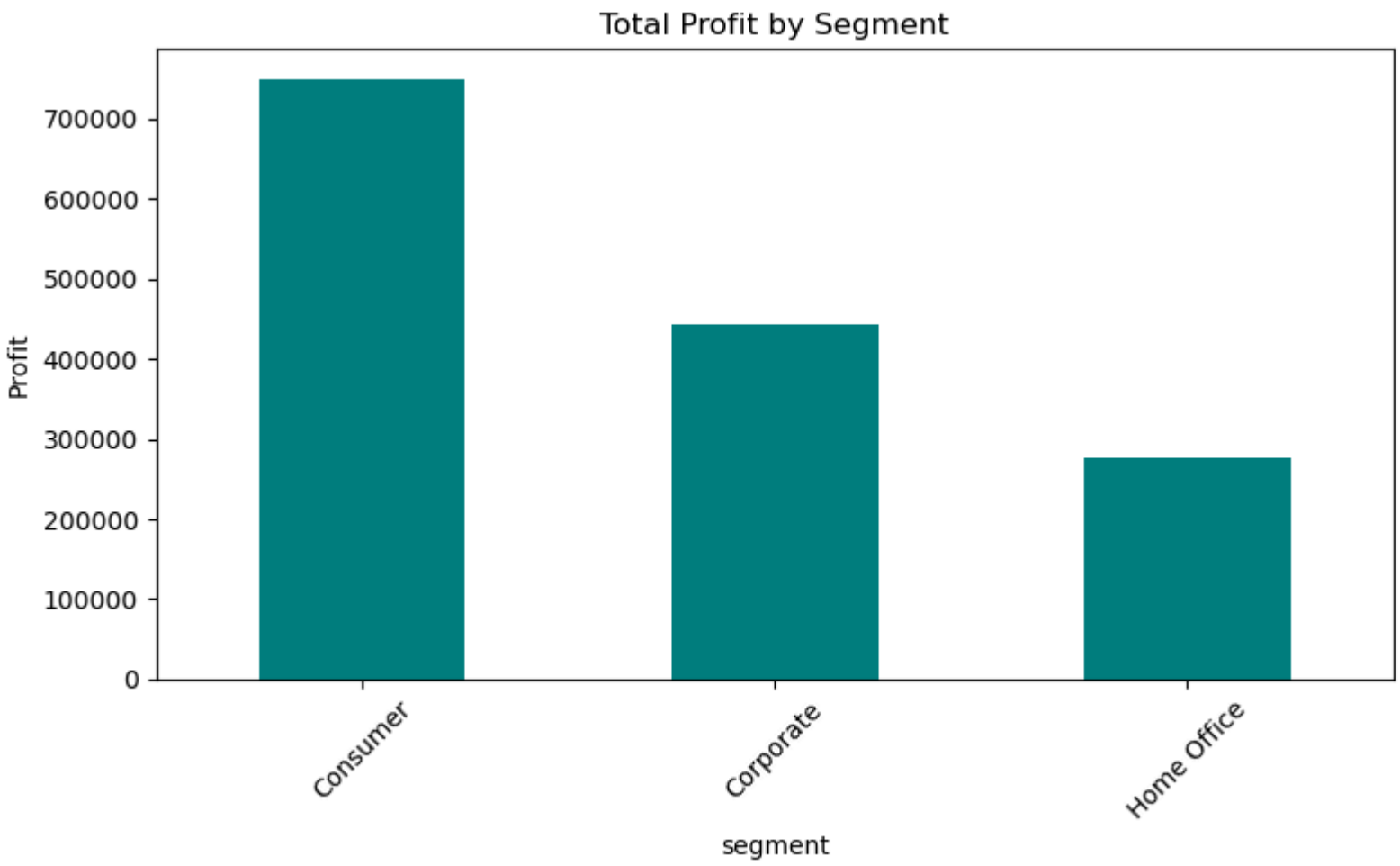
```
top_countries = df.groupby('country')['sales'].sum().nlargest(10)
plt.figure(figsize=(8, 5))
top_countries.plot(kind='barh', color='teal')
plt.title('Top 10 Countries by Sales')
plt.xlabel('Sales')
plt.tight_layout()
plt.show()
```




### Total profit By Segment

In [197…

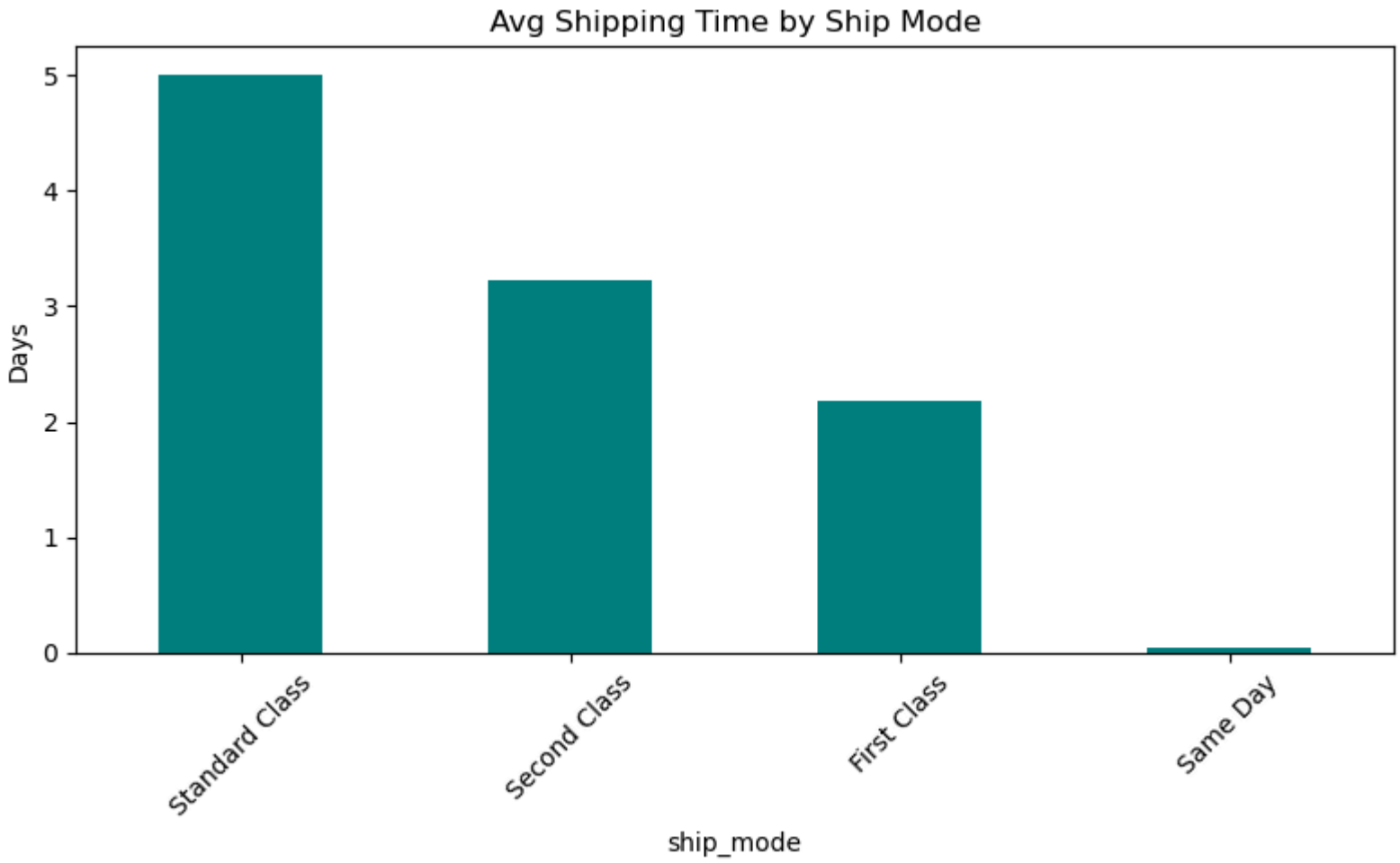
```
segment_profit = df.groupby('segment')['profit'].sum().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
segment_profit.plot(kind='bar', color='teal')
plt.title('Total Profit by Segment')
plt.ylabel('Profit')
plt.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()
```




### Avg Shipping Time by Ship Mode

In [202…

```
ship_time = df.groupby('ship_mode')['days_to_ship'].mean().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
ship_time.plot(kind='bar', color='teal')
plt.title('Avg Shipping Time by Ship Mode')
plt.ylabel('Days')
plt.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()
```




### Avg Shipping Time by Order Priority

In [206…

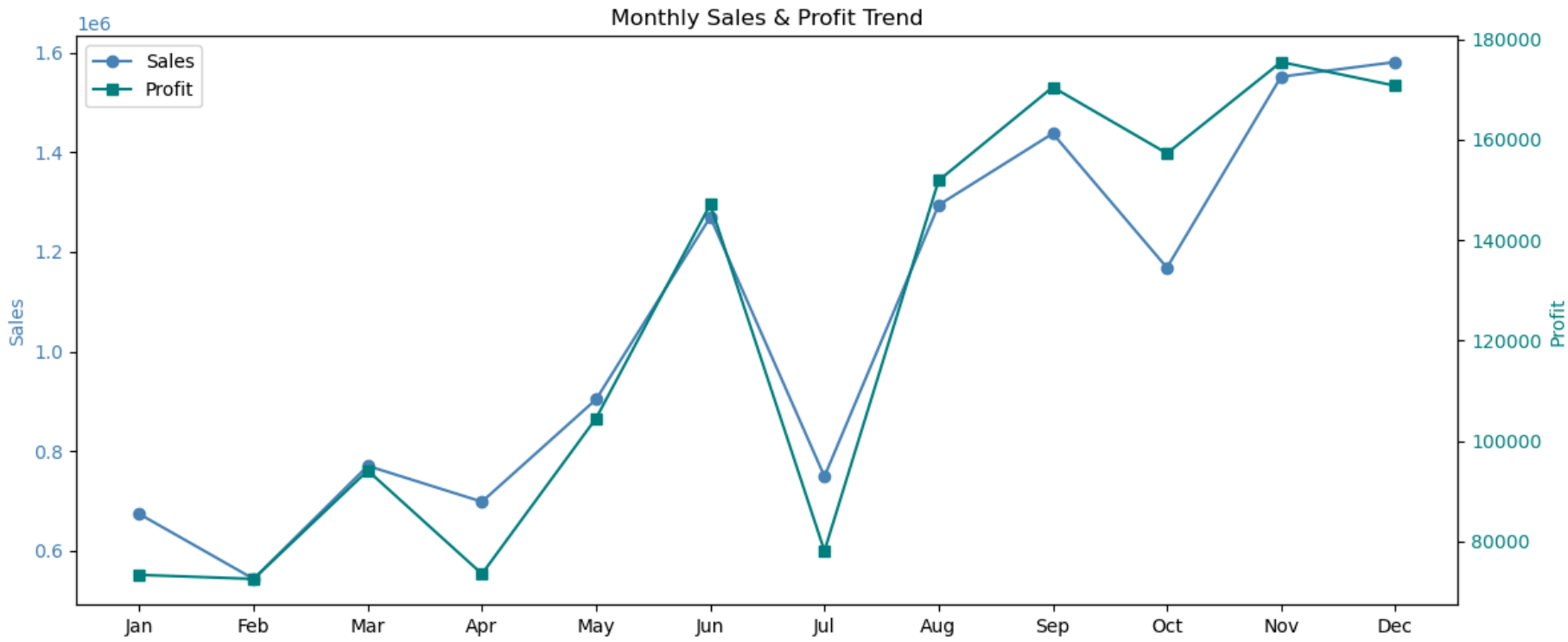
```
priority_time = df.groupby('order_priority')['days_to_ship'].mean().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
priority_time.plot(kind='bar', color='teal')
plt.title('Avg Shipping Time by Order Priority')
plt.ylabel('Days')
plt.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()
```

### Monthly Sales & Profit Trend

In [209…

```
monthly = df.groupby('month').agg(
    sales  = ('sales', 'sum'),
    profit = ('profit', 'sum')
)

fig, ax1 = plt.subplots(figsize=(12, 5))

ax1.plot(monthly.index, monthly['sales'],
         marker='o', color='steelblue', label='Sales')
ax1.set_ylabel('Sales', color='steelblue')
ax1.tick_params(axis='y', labelcolor='steelblue')

ax2 = ax1.twinx()
```

```
ax2.plot(monthly.index, monthly['profit'],
         marker='s', color='teal', label='Profit')
ax2.set_ylabel('Profit', color='teal')
ax2.tick_params(axis='y', labelcolor='teal')


ax1.set_xticks(range(1, 13))
ax1.set_xticklabels(['Jan','Feb','Mar','Apr','May','Jun',
                     'Jul','Aug','Sep','Oct','Nov','Dec'])

lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc='upper left')

plt.title('Monthly Sales & Profit Trend')
plt.tight_layout()
plt.show()
```




In [ ]: